OR-AS Operations Research Applications and Solutions	Case Name: Emergency Department	Sector	Construction (civil)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	■
			Schedule with costs	■
		Risk Analysis	Random simulation	■
			One of nine std. scenarios	
Submitted by	Jens Martens		User defined distributions	
Date	2019	Project Control	Automatic tracking	
File Name	C2019-03 Emergency Department		Tracking based on user input	■

Project authenticity

1. Project description

The construction and installation of an emergency department facility, including structural works, medical systems, and finishing activities.

The project consists of activity, resource, and cost data obtained directly from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	18
Planned Duration (PD)	413d
Budget At Completion (BAC)	967,878.00€
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	37%
Activity Distribution (AD)	43%
Length of Arcs (LA)	3%
Topological Float (TF)	37%

* standard eight-hour working days=

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	14	24	3.41
CRI-rho	20	26	2.27
CRI-tau	25	36	1.70

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	41	51	0.39
SI	49	45	0.23
SSI	12	24	1.95
CRI-r	16	19	1.84
CRI-rho	15	18	1.88
CRI-tau	12	13	1.27

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	N/A	N/A
PV - SPI	N/A	N/A
PV - SCI	N/A	N/A
ED - 1	N/A	N/A
ED - SPI	N/A	N/A
ED - SCI	N/A	N/A
ES - 1	N/A	N/A
ES - SPI(t)	N/A	N/A
ES - SCI(t)	N/A	N/A

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	N/A	N/A
CPI	N/A	N/A
SPI	N/A	N/A
SPI(t)	N/A	N/A
SCI	N/A	N/A
SCI(t)	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 10 tracking periods. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	N/A	N/A	N/A	N/A	N/A	N/A	N/A
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	N/A	N/A	N/A	N/A	N/A	N/A	N/A

2.3.3.2. Time forecasting

PD	413	Real Duration	521	Late	26.2%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	N/A	N/A
PV - SPI	N/A	N/A	N/A	N/A
PV - SCI	N/A	N/A	N/A	N/A
ED - 1	N/A	N/A	N/A	N/A
ED - SPI	N/A	N/A	N/A	N/A
ED - SCI	N/A	N/A	N/A	N/A
ES - 1	N/A	N/A	N/A	N/A
ES - SPI(t)	N/A	N/A	N/A	N/A
ES - SCI(t)	N/A	N/A	N/A	N/A

2.3.3.3. Cost forecasting

BAC	967,878.00€	Real Cost	1,270,875.82€	Over budget	31.3%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	N/A	N/A	N/A	N/A
CPI	N/A	N/A	N/A	N/A
SPI	N/A	N/A	N/A	N/A
SPI(t)	N/A	N/A	N/A	N/A
SCI	N/A	N/A	N/A	N/A
SCI(t)	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A	N/A	N/A